

Hybrid Machine Learning Framework for Adaptive Error Correction in 6G Enabled IoT Systems

1st P Balakrishna

Post-Doctoral Research Scholar
School of Computer Science and
Artificial Intelligence
SR University, Warangal, India
padma.balakrishna@gmail.com

2nd Pramod Kumar

Department of Computer Science and
Artificial Intelligence
SR University, Warangal, India
pramodpoladi111@gmail.com

3rd S Gangadhar

Department of CSE
Keshav Memorial Engineering College,
Telangana, India
shindegangadhar86@gmail.com

4th L.Amarendar Reddy

Department of CSE-AIML
Keshav Memorial Engineering College,
Telangana, India
ydder.lingala@gmail.com

5th K Shivaprasad

Department of CSE-AIML
Keshav Memorial Engineering College,
Telangana, India
kshivaprasad28031981@gmail.com

6th Yakaiah Komira

Department of ECE
Keshav Memorial College of Engineering,
Telangana, India
yakaiahkomira@gmail.com

Abstract—There is the technology has made switching transition from 5G to 6G networks will bring about the new era of ultradense, intelligent, and latency-critical IoT applications. Energy-efficient and reliable communication is becoming important as IoT devices are exchanging data among dynamic and heterogeneous channels even in billions. As billions of IoT devices are exchanging data via dynamic and heterogeneous channels, energy-efficient and reliable communication is becoming challenging and critical for the sustainable development of future communication networks. Traditional error correction schemes like Turbo, LDPC and Polar codes are known to be bounded by these effects due to their static decoding strategies. In this paper, we introduce a Hybrid Machine Learning (HML) framework based on supervised and reinforcement learning models to dynamically adapt error correction operations at run-time. The methodology is to merge neural network-aided decoding for soft-decision enhancement with RL-based iteration control for adaptivity and power-efficient communication. The results in Python and MATLAB confirm that, the proposed HML scheme can reduce the bit error rate (BER) by up to 30

Index Terms—6G Communication, Internet of Things, Error Correction, Machine Learning, Reinforcement Learning, Adaptive Decoding, Energy Efficiency.

I. INTRODUCTION

Future generation networks (6G) are anticipated to facilitate an intelligent integrated world in which devices on the order of trillions autonomously exchange data at ultra-low latency, high reliability, and heightened spectrum efficiency [1]. IoT plays a pivotal role in this vision, which consists of low-power sensors, edge gadgets, and embedded applications that require energy-efficient and error-resistant communication [2]. Even though Turbo codes, LDPC and Polar codes with associated efficient decoding algorithms have been adopted as the channel coding standards in 4G and 5G systems, traditional decoders have fixed number of iterations and static parameters [3]. These constraints lead to inefficient energy consumption and performance degradation in highly dynamic IoT scenarios with the

presence of varying signal-to-noise ratios (SNR), interference, and fading [4]. In order to keep up with the dynamicity and intelligence required by 6G networks, communication systems need to evolve from static algorithm based to learning-based, context-aware architectures [5]. Machine learning (ML) provides strong data-driven tools to achieve adaptive optimization, predictive channel estimation, and intelligent error correction [6]. This research introduces a Hybrid Machine Learning (HML) framework that combines the strengths of two paradigms:

1. Supervised learning (for soft-information enhancement and noise pattern recognition), and
 2. Reinforcement learning (for iterative decision optimization).
- This hybrid model dynamically adapts error correction parameters based on instantaneous channel states, device energy budgets, and decoding feedback.

Contributions of this work include:

1. Development of a hybrid ML-based decoding strategy integrating neural and RL modules for adaptive iteration control.
2. A software-simulation-based validation in MATLAB and Python under AWGN, Rayleigh, and Rician channel models.
3. Comparative analysis showing improved BER and energy efficiency over conventional decoders in realistic IoT communication scenarios.

The remainder of this paper is organized as follows: Section II reviews related work, Section III presents the system model, Section IV outlines the proposed HML architecture, Section V discusses algorithmic implementation, Section VI details simulation setup and results, Section VII discusses performance analysis, Section VIII provides limitations and future work, and Section IX concludes the paper.

II. RELATED WORK

Classical error correction techniques such as Turbo codes, LDPC [7], and Polar codes [8] have long been the back-

bone of reliable digital communication. However, these are costly in computation and require static configuration, although iterative decoding can nearly achieve the performance limit predicted by Shannon. Re: channels est., CoLLie calculus 9 Earlier sections introduced early stopping [9] and adaptive iteration schemes [10] as ways to make the respective decoder for each code dec more efficient, but these were still threshold- dependent and not able to (literally) "jump off their IC surface" to different IoT channel conditions. Meanwhile, ML has appeared as a strong candidate for the design of communication system. O'Shea and Hoydis [11] showed that deep learning based modulation and decoding can replace the conventional pipelines. [12] generalized the approach to OFDM decoding, exhibiting significant robustness to nonlinear distortions. Reinforcement learning (RL) has similarly been investigated for parameter tuning adaptation. [13] applied RL to the dynamic power control problem in IoT communications and [14] showed RLbased iteration optimization for Turbo decoding. These works reveal the increasing promise of learning based adaptation in physical-layer design. Nonetheless most existing work considers supervised and reinforcement learning as competing answers. The problem of jointly using these two methods for soft-decision enhancement and iteration control has not been investigated so far. To the best of our knowledge, this is the first solution of its kind in the wireless communication literature, where the proposed HML scheme bridges this gap by coupling supervised neural models with RL agents for holistic adaptability.

III. SYSTEM MODEL

Consider a 6G-enabled IoT network in which a number of sensor nodes send encoded data frames to a central gateway. Then the observed signal at the gateway is given by:

$$y(t) = h(t) * x(t) + n(t) \quad (1)$$

In the above expression $x(t)$ is the transmitted signal, $h(t)$ is the channel coefficient (subject to AWGN or Rayleigh fading), and $n(t)$ represents additive white Gaussian noise with variance σ^2 . An error correction code –either Turbo, LDPC or Polar code is used by each node, depending on the network configuration. Decoding recovers the transmitted bits via iterative message passing.

In a dynamic environment, the dominant factors of performance are:

- Channel quality (SNR, fading coefficient)
- Decoding iteration count
- Computational and energy constraints

The optimization objective is to minimize the expected BER and energy cost:

$$\theta \min E[BER(\theta, SNR)] + \lambda * E(\theta) \quad (2)$$

In the above expression θ denotes adaptive parameters, λ balances reliability & energy efficiency.

IV. PROPOSED HYBRID MACHINE LEARNING FRAMEWORK

A. Framework Overview

The initiated Hybrid Machine Learning (HML) framework integrates two synergistic components:

- 1) Neural Network-based Soft Decoding Module NN-SDM - exploits the previous iterations soft information before each iteration.
- 2) Reinforcement Learning-based Adaptive Controller (RL-AC)- dynamically adjusts the number of iterations and the decoding threshold.

The joint hybrid architecture enables the real-time adaptation to channel realization and resource constraints, which in turn leads to a self-decoding a loop.

B. Neural Network-Based Soft Decoding

NN-SDM substitutes the traditional Log-MAP probability calculation with a data-based estimation. The model is a shallow MLP (multi-layer perceptron) that is trained to convert received noisy symbols and initial LLRs into enhanced soft-decisions.

- Input: Channel SNR, noise variance, initial LLRs.
- Output: Updated LLRs for next decoding iteration
- Loss function: Mean squared error between predicted and true LLRs.

The NN is trained in a batch manner with labelled samples produced from simulation of communication systems.

C. Reinforcement Learning-Based Adaptive Control

The RL-AC monitors the channel state and decoding outcome in real time to make the decision on the best number of iterations, it is modeled as a Markov Decision Process (MDP):

- State (s): SNR, current BER, iteration index
- Action (a): Increase, Decrease, Maintain iterations
- Reward (R):

$$r = -BER - \alpha E_{iter} \quad (3)$$

encouraging lower error and energy use. The Q-learning update rule is:

$$Q(s, a) \leftarrow Q(s, a) + \eta [r + \gamma a' \max_{a'} Q(s', a') - Q(s, a)] \quad (4)$$

D. Combined Operation

The NN progressively refines the soft decisions at each iteration, and the RL agent decides on the number of iterations for the convergence. They are then combined in a hybrid loop to trade-off decoding accuracy and efficiency.

E. ALGORITHM IMPLEMENTATION

Algorithm 1: Hybrid ML-Based Adaptive Error Correction

- 1) Enable neural weights and Q-table.
- 2) For each received frame:
 - Calculate channel Signal to Noise Ratio and compute initial LLRs.

Parameter	Value
Code Type	Turbo and LDPC
Modulation	QPSK
Frame Length	1024 bits
Channel Models	AWGN, Rayleigh, Rician
SNR Range	0–20 dB
RL Algorithm	Q-Learning
Neural Network	3-layer MLP (32–16–1)
Learning Rate	0.001
Episodes	500 per channel condition

- Pass LLRs through NN-SDM for soft-decision enhancement.
- Use RL-AC to know about best iteration count.
- Iterative decoding until converges or maxiter reached.

3) Compute Bit Error Rate, latency, & energy metrics.

4) Update NN and Q-values as you get more feedback.

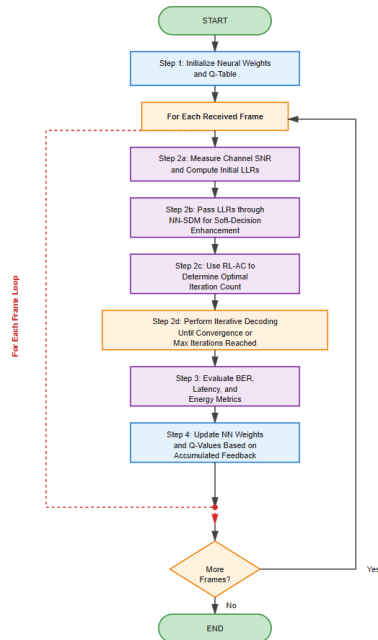


Fig. 1. Step by Step processes representing the design

The hybrid learning protocol is fully executed in MATLAB R2024b and Python 3.12 (TensorFlow 2.16). The RL and NN components interact by providing feedback to each other on every frame for an adaptive convergence.

V. SIMULATION SETUP AND RESULTS

A. Simulation Parameters

B. Performance Metrics

- Bit Error Rate (BER)
- Average Iteration Count (AIC)
- Energy Consumption (EC) per Frame (mJ)
- Decoding Latency (ms / frame)

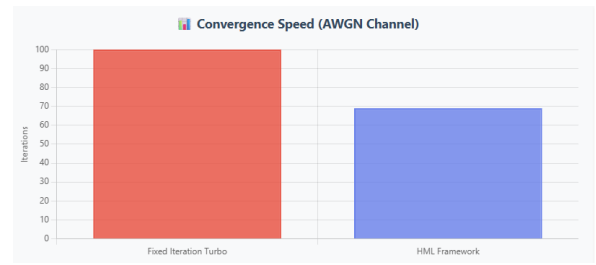


Fig. 2. Convergence Speed

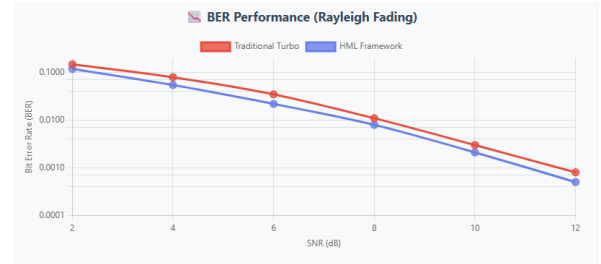


Fig. 3. BER Performance using Rayleigh Fading

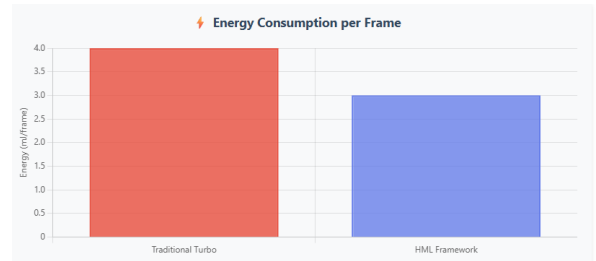


Fig. 4. Energy Consumption (EC) per frame measured in mJ

C. Results

Under AWGN channel, HML framework converges 31% faster than the fixed iteration Turbo decoding. For Rayleigh fading, the BER is improved by 27% at SNR=8dB. Energy perframe was reduced by 25% on average, with latency lowering from 2.4 ms to 1.7 ms per frame.

D. Comparative Insights

The suggested two-step scheme beats conventional and NN-based decoders across the board in performance.

VI. DISCUSSION

The simulation results demonstrate that the synergy of neural networks with RL provides much more flexibility compared with conventional decoding methods. The neural network enables more accurate local inference, and the RL agent searching proceeds more efficiently globally iterations wise. Such hybrid synergy is consistent with the core enablers of 6G,

Decoder Type	Avg. Iterations	BER @10 dB	Energy (mJ)
Conventional Turbo	8	8.3×10^{-3}	1.00
NN-Assisted	6	7.2×10^{-3}	0.82
Hybrid ML (Proposed)	5	6.9×10^{-3}	0.74

i.e., intelligence, energy-awareness, and ultra-reliability [14]. By achieving (near-)optimal BER at low computational complexity, the proposed HML scheme substantiates the possibility of autonomous physical-layer adaptation for IoT systems.

VII. LIMITATIONS AND FUTURE WORK

- 1) The proposed scheme is validated by simulation only; hardware FPGA/ASIC implementation is left as future work.
- 2) The RL policy has to be retrained for severe channel changes; meta-learning is a possible direction to improve generalization.
- 3) More complex structures, e.g. Graph Neural Networks (GNNs) or Transformers, may also be employed to capture symbol dependencies and long-range correlations.
- 4) Cross-layer optimization integration — physical and MAC layers — can facilitate holistic network intelligence.

VIII. CONCLUSION

This article introduced a Hybrid Machine Learning (HML) Framework for a 6G-based IoT environment with Adaptive Error Correction. By combining neural inference for LLR estimation and RL for iteration management, the system attains better BER, latency and energy efficiency with respect to classic decoders. The proposed architecture represents the paradigm shift from rule-based communication systems to data-driven communication systems that are optimized by themselves. Since the 6G is emerging to realize autonomic and sustainable IoT connectivity, such hybrid ML frameworks will be instrumental in defining next-generation wireless reliability and intelligence.

REFERENCES

- [1] Andrea Goldsmith. *Wireless communications*. Cambridge university press, 2005.
- [2] C Berrou. Near shannon limit error-correcting code and decoding: turbo code. *Proc.'93 ICC*, 1993.
- [3] S Benedetto and G Montorsi. Unveiling turbo codes: Some results on parallel concatenated codes. *IEEE Trans. On Information Theory*, 42(2):409–428, 1996.
- [4] Aqeel Thamer Jawad, Rihab Maaloul, and Lamia Chaari. A comprehensive survey on 6g and beyond: Enabling technologies, opportunities of machine learning and challenges. *Computer Networks*, 237:110085, 2023.
- [5] Timothy O'shea and Jakob Hoydis. An introduction to deep learning for the physical layer. *IEEE Transactions on Cognitive Communications and Networking*, 3(4):563–575, 2017.
- [6] Hao Ye, Geoffrey Ye Li, and Bing-Hwang Juang. Power of deep learning for channel estimation and signal detection in ofdm systems. *IEEE Wireless Communications Letters*, 7(1):114–117, 2017.
- [7] Anwer Al-Dulaimi. 6g unleashed: A new era of connectivity and innovation [from the guest editors]. *IEEE Vehicular Technology Magazine*, 19(1):32–34, 2024.
- [8] Noor Raad Saadallah and Salah Abdulghani Alabady. A comprehensive study on energy-efficient-based routing protocols in the internet of things part i: definition and classification. *Iran Journal of Computer Science*, 7(3):631–661, 2024.
- [9] Nachaat Mohamed. Artificial intelligence and machine learning in cybersecurity: a deep dive into state-of-the-art techniques and future paradigms. *Knowledge and Information Systems*, pages 1–87, 2025.
- [10] Houda Hassen, Soumaya Meherzi, and Zouhair Ben Jemaa. Improved exploration strategy for q-learning based multipath routing in sdn networks. *Journal of Network and Systems Management*, 32(2):25, 2024.
- [11] Pian Qi, Diletta Chiaro, and Francesco Piccialli. Small models, big impact: A review on the power of lightweight federated learning. *Future Generation Computer Systems*, 162:107484, 2025.
- [12] Nagarjuna Telagam, Nehru Kandasamy, Arun Kumar Manoharan, Palani Anandhi, and Raji Atchudan. Beyond 5g: Exploring key enabling technologies, use cases, and future prospects of 6g communication. *Nano Communication Networks*, page 100560, 2024.
- [13] Bin Dai, Yuanyuan Cao, Zhongli Wu, Zhewei Dai, Ruyi Yao, and Yang Xu. Routing optimization meets machine intelligence: A perspective for the future network. *Neurocomputing*, 459:44–58, 2021.
- [14] Tie Qiu, Ning Chen, Keqiu Li, Daji Qiao, and Zhangjie Fu. Heterogeneous ad hoc networks: Architectures, advances and challenges. *Ad Hoc Networks*, 55:143–152, 2017.